

React Todo List with API Integration

Role: Software Developer

Difficulty: junior

Project Type: frontend

Duration: 3 days

Focus: frontend

Candidate: Rahul Sharma

Project Overview

Create a React app that fetches a todo list from a public API, allows CRUD operations locally, and persists data in localStorage.

Technologies

- React

Requirements

- Use React to build UI
- Fetch data from JSONPlaceholder
- Implement add/edit/delete of todos
- Persist todos in localStorage
- Use functional components and hooks
- Write basic unit tests

Objectives

- Build a functional todo app
- Demonstrate API integration
- Show state management skills
- Ensure data persistence
- Apply clean code practices

Functional Requirements

- Display list of todos
- Add new todo
- Edit existing todo
- Delete todo
- Persist todos in localStorage
- Show loading indicator during fetch

Non-Functional Requirements

- Responsive layout
- Fast load time
- Accessible UI
- Error handling for API failures
- Maintainable code structure

Modules

UI

Components and styling

- TodoList
- TodoItem
- TodoForm
- LoadingSpinner

DataService

API interactions

- fetchTodos
- addTodo
- updateTodo
- deleteTodo

Storage

LocalStorage persistence

- saveTodos
- loadTodos
- clearTodos

Implementation Plan

Setup

- Create React app
- Add dependencies
- Set up folder structure

Development

- Build UI components
- Implement DataService
- Integrate localStorage
- Add routing if needed

Testing & Polish

- Write unit tests
- Fix bugs
- Optimize performance
- Prepare README

Evaluation Criteria

Code Quality (30%)

Clean, readable, and modular code

Functionality (30%)

All features work as specified

UI/UX (20%)

Responsive and user-friendly interface

Performance (20%)

Fast load and minimal lag

Deliverables

- Source code on GitHub
- README with setup instructions
- Demo video or GIF
- Unit test coverage report

Suggested Folder Structure

- src/components
- src/services
- src/utils
- src/tests

API Endpoints

GET <https://jsonplaceholder.typicode.com/todos>

Fetch initial todo list